Oracle MCP Server Configuration with Claude Code in VS Code

A complete step-by-step guide to configure the Oracle SQLcl MCP server with Claude Code (Cline) in Visual Studio Code.

Prerequisites

Before you start, ensure you have:

1. **Java Development Kit (JDK)**
   - JDK 17 or higher
   - Verify: `java -version`
   - Set `JAVA_HOME` environment variable
2. **Visual Studio Code**
   - Latest version installed
   - Download from: https://code.visualstudio.com/
3. **Oracle Database Access**
   - Active Oracle Database connection
   - Database username and password
   - Database connection details (host, port, service name)
4. **An MCP-compatible client**
   - Claude Code (Cline) for VS Code (recommended)
   - Or GitHub Copilot in VS Code

Step 1: Download and Install SQLcl

SQLcl is Oracle's command-line tool that includes the MCP server functionality.

On Windows:

1. Download SQLcl from: https://www.oracle.com/database/sqldeveloper/technologies/sqlcl/download/
2. Unzip to a folder (e.g., `C:\app\sqlcl\`)
3. Set up environment variables:

```
PATH: C:\app\sqlcl\bin
JAVA_HOME: C:\Program Files\Java\jdk-17 (or your JDK location)
```

4. Verify installation:

```cmd
sql -version
```

On macOS/Linux:

1. Download SQLcl from the same URL
2. Unzip to a folder (e.g., `/opt/sqlcl/`)
3. Set environment variables in your shell profile:

```bash
export JAVA_HOME=/usr/libexec/java_home -v 17
export PATH="/opt/sqlcl/bin:$PATH"
```

4. Make the script executable:

```bash
chmod +x /opt/sqlcl/bin/sql
```

5. Verify installation:

```bash
sql -version
```

Step 2: Create and Save an Oracle Database Connection

The SQLcl MCP server uses named, saved connections with passwords.

Using SQLcl Command Line:

1. Open a terminal and start SQLcl:

```
sql /nolog
```

2. Create a saved connection with password (example for HR user):

```sql
conn -save claude_hr -savepwd hr@(DESCRIPTION=(ADDRESS_LIST=(ADDRESS=(PROTOCOL=TC
```

3. When prompted, enter the password and confirm
4. Verify the connection:

```sql
conn claude_hr
select * from user_tables;
exit
```

Connection Details Explained:

- `-save claude_hr`: Names your connection "claude_hr" (use a meaningful name)
- `-savepwd`: Saves the password securely
- `hr`: Database username
- `your_host`: Database server hostname or IP
- `1521`: Default Oracle database port (change if different)
- `your_service_name`: Oracle service name

Step 3: Install Claude Code (Cline) Extension

If you haven't already installed Cline in VS Code:

1. Open VS Code
2. Click **Extensions** on the left sidebar (or press `Ctrl+Shift+X`)
3. Search for **"Cline"** (by Cline)
4. Click **Install**
5. Click **Trust Publisher** and **Install** when prompted
6. The extension appears on the sidebar

Step 4: Configure MCP Server in Claude Code (Option 1 - Command Line)

Option 1A: Using `claude mcp add` command

1. Open a terminal/command prompt
2. Run this command to add the SQLcl MCP server: **On Windows:**

```cmd
claude mcp add --transport stdio --scope user sqlcl -- "c:\app\sqlcl\bin\sql.exe"
```

On macOS/Linux:

```bash
claude mcp add --transport stdio --scope user sqlcl -- /opt/sqlcl/bin/sql -mcp
```

Replace paths with your actual SQLcl installation paths.

3. Verify the configuration was added:

```
claude mcp list
```

Step 4: Configure MCP Server in Claude Code (Option 2 - JSON)

Option 2A: Using `claude mcp add-json` command

1. Open a terminal/command prompt
2. Run this command: **On Windows:**

```cmd
claude mcp add-json sqlcl "{\"command\": \"c:\\app\\sqlcl\\bin\\sql.exe\", \"args
```

On macOS/Linux:

```bash
claude mcp add-json sqlcl '{"command": "/opt/sqlcl/bin/sql", "args": ["-mcp"]}'
```

Step 5: Configure MCP Server in Claude Code (Option 3 - Config File)

Option 3A: Manual Configuration File Edit

Claude Code stores MCP configurations in a settings file.

1. Find the configuration file:
   - **Windows:** `%APPDATA%\Cline\cline_mcp_settings.json` or `~/.cline/cline_mcp_settings.json`
   - **macOS:** `~/.cline/cline_mcp_settings.json`
   - **Linux:** `~/.cline/cline_mcp_settings.json`
2. If the file doesn't exist, create it with this content: **On Windows:**

```json
{
  "mcpServers": {
    "sqlcl": {
      "command": "c:\\app\\sqlcl\\bin\\sql.exe",
      "args": ["-mcp"],
      "disabled": false
    }
  }
}
```

On macOS/Linux:

```json
{
  "mcpServers": {
    "sqlcl": {
      "command": "/opt/sqlcl/bin/sql",
      "args": ["-mcp"],
      "disabled": false
    }
  }
}
```

3. Save the file (Ctrl+S)

Step 6: Verify MCP Server Configuration in VS Code

1. Open VS Code
2. Open the Cline extension (click the icon on the sidebar)
3. Look for **MCP Servers** section
4. Click on it to view configured servers
5. You should see **sqlcl** listed with status showing it's enabled
6. If not visible, try restarting VS Code:
    - Close VS Code completely
    - Reopen it

Step 7: Test the Oracle MCP Connection

1. In VS Code, open the Cline chat panel
2. In the message input field, ask Cline a database question:

```
What tables are in my database? Show me the schema for the EMPLOYEES table.
```

3. Cline should:
    - Invoke the SQLcl MCP server tools
    - Connect to your saved Oracle database connection
    - Retrieve and display the information
4. Example response:

```
The schema for the EMPLOYEES table includes:
- EMPLOYEE_ID (NUMBER)
- FIRST_NAME (VARCHAR2)
- LAST_NAME (VARCHAR2)
- EMAIL (VARCHAR2)
...
```

Step 8: Available Oracle MCP Tools

Once configured, Cline has access to these SQLcl MCP tools:

1. **run-sql** - Execute SQL queries and PL/SQL blocks

- Example: "Run a query to find all employees in department 10"

2. **get_database_info** - Retrieve general database information
   - Example: "What database version are we running?"
3. **get_table_schema** - Get detailed schema for a specific table
   - Example: "Show me the columns and data types for the DEPARTMENTS table"
4. **get_tables_schema** - Get schema for multiple tables at once
   - Example: "Show schemas for EMPLOYEES and DEPARTMENTS"
5. **search_tables** - Search for tables by name pattern
   - Example: "Find all tables starting with 'EMP'"

Step 9: Advanced Configuration (Optional)

Add Environment Variables:

If your Oracle setup requires specific environment variables:

On Windows (add to cline_mcp_settings.json):

```json
{
  "mcpServers": {
    "sqlcl": {
      "command": "c:\\app\\sqlcl\\bin\\sql.exe",
      "args": ["-mcp"],
      "disabled": false,
      "env": {
        "ORACLE_HOME": "c:\\app\\oracle",
        "TNS_ADMIN": "c:\\app\\oracle\\network\\admin"
      }
    }
  }
}
```

On macOS/Linux:

```json
```

```
{
  "mcpServers": {
    "sqlcl": {
      "command": "/opt/sqlcl/bin/sql",
      "args": ["-mcp"],
      "disabled": false,
      "env": {
        "ORACLE_HOME": "/opt/oracle",
        "TNS_ADMIN": "/opt/oracle/network/admin"
      }
    }
  }
}
```

Multiple Database Connections:

To work with multiple Oracle databases, create multiple named connections in SQLcl:

```sql
conn -save db_prod -savepwd produser@prod_host:1521/prod_service
conn -save db_test -savepwd testuser@test_host:1521/test_service
```

Then use Cline to specify which connection to use in your questions.

Troubleshooting

Issue: "Command not found: sql"

- **Solution:** Ensure SQLcl is in your PATH and JAVA_HOME is set correctly
- Verify: `echo $PATH` (macOS/Linux) or `echo %PATH%` (Windows)

Issue: "Connection refused" when testing

- **Solution:**
  - Verify your Oracle database is running
  - Check database hostname, port, and service name
  - Confirm the saved connection works: `conn claude_hr` in SQLcl

Issue: MCP Server not appearing in Cline

- **Solution**:
  - Restart VS Code completely
  - Check the config file for JSON syntax errors
  - Verify the SQLcl path is correct and executable

Issue: "Password required" when connecting

- **Solution**:
  - Reconnect your saved connection using `-savepwd` flag
  - Delete and recreate the saved connection

Issue: SQLcl version compatibility

- **Note**: SQLcl version 25.4+ has some issues with certain MCP clients
- **Recommendation**: Use version 25.3.2.317.1117 if you encounter crashes
- Download from: https://www.oracle.com/database/sqldeveloper/technologies/sqlcl/download/

Next Steps

1. **Explore database structure**: Ask Cline about your schema
2. **Generate SQL**: Let Cline write and execute queries
3. **Analyze data**: Ask for insights from your database
4. **Debug performance**: Use Cline to identify slow queries

Example prompts:

- "Write a query to find the top 10 customers by revenue"
- "Help me optimize this slow query: [paste query]"
- "What's the relationship between the ORDERS and CUSTOMERS tables?"
- "Create an index on the EMAIL column of the USERS table"

References

- SQLcl Documentation

- Oracle MCP Server Guide
- Claude Code Repository
- Model Context Protocol (MCP)

Support

If you encounter issues:

1. Check SQLcl logs for error messages
2. Test your database connection manually in SQLcl
3. Verify all paths and environment variables are correct
4. Consult the Oracle documentation links above
5. Check the Claude/Cline community forums

Last Updated: May 2026
Tested with: SQLcl 25.3, VS Code, Cline MCP Support